

# PARUL DHINGRA

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## EDUCATION

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### Indian Institute of Remote Sensing, ISRO

2019 – 2020

*Post-graduate Diploma in Remote Sensing and GIS (Grade: 8.54/10.00 with distinction)*

Dehradun, Uttarakhand

### Friedrich-Alexander University (FAU), Erlangen-Nuremberg

2015 – 2018

*M.Sc. in Communications and Multimedia Engineering (Grade: 2.1/5.0 wherein 1.0 is highest)*

Erlangen, Germany

### Manipal Institute of Technology (MIT), Manipal

2010 – 2014

*B.E. in Electronics and Communication Engineering (Grade: 8.52/10.00)*

Manipal, Karnataka

### Cambridge Court High School, CBSE

2010

*Class XII (Percentage: 88.8 % (Scores - Physics: 88/100, Chemistry: 94/100, Maths: 93/100))*

Jaipur, Rajasthan

### Saint Mary's Senior Sec. School, ICSE

2008

*Class X (Percentage: 93.42 % (Scores - Science: 94/100, Maths: 92/100))*

Sikar, Rajasthan

## RESEARCH EXPERIENCE

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### Indian Institute of Remote Sensing, Dehradun

May 2020 – Aug 2020

*Post-graduate Diploma Trainee*

Dehradun, Uttarakhand

- Title: Automated target feature extraction using ultra-high spatial resolution images
- Supervisor: Dr. Hina Pande, Photogrammetry and Remote Sensing Department
- The objective of project was to propose a computer vision method for automated target extraction (airplane feature) from ultra-high spatial resolution data images (source of data: UAVs), which consist of several background features like buildings, vehicles, roads etc., using deep learning networks. Annotated data images obtained from eBee Classic drones. Implemented and trained deep neural network architecture for object detection and semantic segmentation for extraction of target feature. Implemented cloud-computing with Microsoft Azure notebook and Google Colab.

### KTH Royal Institute of Technology, Stockholm

Sept 2017 – Feb 2018

*Master's Thesis Intern*

Stockholm, Sweden

- Title: Framing vocoder for data compression using deep learning
- Supervisor: Prof. Jonas Beskow, Division of Speech, Music and Hearing
- Researched and implemented novel algorithm for vocoders, which incorporated building deep learning network for autoencoders with extensive research on speech signals. Developed a framework for pre-processing speech signals and prepared the dataset for machine learning. Analyzed the framework with subjective audio listening tests on CrowdFlower platform. Implemented deep learning neural networks on GeForce-GTX 1080 GPU in Keras-TensorFlow backend environment. Programmed in MATLAB and Python.
- Grade: 1.3/5.0 (1.0 is the highest.)

### Fraunhofer IIS, Erlangen

April 2017 – July 2017

*Research Intern*

Erlangen, Bavaria

- Title: Inter-channel Time Difference handling for low delay stereo coding
- Supervisor: Dr. rer. nat. Jan B  the
- Assisted in handling inter-channel time differences and conducting quality tests- MUSHRA and POLQA.

## ACADEMIC TEACHING EXPERIENCE

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### Multimedia Communications Lab, FAU Erlangen-Nuremberg

April 2017 – July 2017

*Tutor*

Erlangen, Bavaria

- Lab course on implementing Video Codecs (video compression and decompression framework).
- Assessed lab assignments, explained concepts to build video codec, and assisted students in writing and debugging the code in MATLAB.

## CONFERENCES AND WORKSHOPS ATTENDED

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- Virtually attended Computer Vision and Pattern Recognition (CVPR) held at Nashville in June 2025. [Link](#)
- Attended workshop on Quantum Communications and Information Processing in National Conference on Communications at IIT-Delhi in March 2025. [Link](#)
- Presented in International Conference On Unmanned Aerial Systems In Geomatics, IIT-Roorkee, April 2021. [Link](#)
- Presented in TRYST - Technical Festival, IIT-Delhi, March 2013. Topic: Green Power Generation in Mobile Phones and Laptops. Objective to present an idea to generate green energy used to charge battery in mobile phones and laptops.
- Presented in International Conference on Electronic Design and Signal Processing, Manipal Institute of Technology, Manipal, Dec 2012.

## PUBLICATIONS

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- Parul Dhingra, Hina Pande, Poonam S. Tiwari and Shefali Agrawal. A Deep Learning Approach for Detection and Segmentation of Airplanes in Ultrahigh-Spatial-Resolution UAV Dataset. In: Jain, K., Mishra, V., Pradhan, B. (eds.) Proceedings of UASG 2021: Wings 4 Sustainability. Lecture Notes in Civil Engineering, vol 304. Springer, Cham, March 2023. [Link](#)
- Parul Dhingra, Jhila Biswas, Anjushree Prasad and Sukanya S Meher. Article: Energy Harvesting using Piezoelectric Materials. International Journal of Computer Applications Special Issue on ICEDSP Volume: ICEDSP(Number 4) : Pages 38-42, ISBN: 973-93-80872-70-8, February 2013. [Link](#)

## OTHER PROFESSIONAL EXPERIENCE

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### M4Com

May 2024 – July 2024

*Junior Software Developer*

*Salem, Baden-Württemberg*

- Assigned project: MAuS. Registered ticket in JIRA for software improvement. Skills: Python and JIRA.
- Submitted resignation due to inter-state and city inconvenient location constraints. Relieved effortlessly with NOC.

### Training in Embedded System Design and Robotics, Hewlett Packard (HP)

June 2013

*Student Intern*

*Jaipur, Rajasthan*

- Project: Obstacle Avoiding Robot
- Designed a four-wheeled robot that moves along the straight path and turns to right or left on detecting an object in its path. Implemented instructions in C and worked with AVR ATmega16 microcontroller.

### Bharat Sanchar Nigam Limited (BSNL)

May 2012 – June 2012

*Student Intern*

*Jaipur, Rajasthan*

- Completed 6 weeks training in Advanced Telecom with Specialization in Data Communication.
- Theory and practical sessions on Telecommunication network, Digital Switching, Fiber Optics Communication, Broadband, Mobile Communication, Intelligent Network, Next Generation Network.
- Received Performance Excellence Certificate.

## AWARDS AND ACHIEVEMENTS

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### Golden Jubilee Fellowship

2019 – 2020

*Indian Institute of Remote Sensing, Dehradun*

- Merit based scholarship awarded to the first rankers for excelling at the institute. Amount: INR 90,000

### Erasmus+ Scholarship

2017 – 2018

*Friedrich-Alexander University, Erlangen-Nuremberg*

- Financial scholarship while working on master's thesis at KTH, Stockholm.

### Teaching Evaluation - Bestes Praktikum, Multimedia Communications Lab

2017

*Friedrich-Alexander University, Erlangen-Nuremberg*

- The lab course, supervised by 5 tutors and supervisor, was ranked 1st out of 45 lab courses in teaching evaluation at Technical Faculty, during summer semester 2017. [Link](#)

### Qualified GATE

2014

### Queensland University of Technology, Brisbane, Australia

2014

*Manipal Institute of Technology, Manipal*

- Due to excellence in academics and co-curricular activities, among the two students awarded funded last semester's degree completion research project scholarship internship, by Manipal Institute of Technology, Manipal University for the duration of 4 months, at The Queensland University of Technology, Brisbane, Australia. Unfortunately, visa process was taking time and our Bachelor Degree award might got delayed due to that. Hence, took the degree completion project in Manipal Institute of Technology, Manipal, during the last semester and completed the project successfully.

### MIT, Manipal Entrance Exam

2010

*Test Center, Jaipur*

- All India Rank 1342 among 15000+ candidates

## SKILLS

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*Languages:* Hindi (Native), English (Fluent), German (B2)

*Programming:* MATLAB, Python

*Others:* Microsoft Office, LaTeX

## COURSEWORK

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- Engineering Mathematics
- Signals and Systems
- Digital Signal Processing
- Pattern Recognition
- Multi-dimensional Signals and Systems
- Speech and Audio Signal Processing
- Image and Video Compression
- Transforms in Signal Processing
- Statistical Signal Processing
- And more.

## EXTRA-CURRICULAR ACTIVITIES/ MEMBERSHIPS/ RESPONSIBILITIES

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- Registered **tutor for Maths and Physics** subjects - IB Diploma at Studienkreis GmbH, Erlangen, 2023 - 2024.
- Qualified **German Language - B2 Exam** at AFI - Language School, Erlangen in Dec 2023.
- **Participation in Bayerisches Rotes Kreuz- Bereitschaft Erlangen 1** (Bavaria Red Cross) events, Sept 2023.
- **Volunteer (UNICEF-Hochschulgruppe-Nürnberg (2016-2018))**. Assistance in charity run and booth set-ups at various events in school, parks, career fair, etc. for raising awareness about UNICEF, children's games, and donations.
- **Student Volunteer (Interspeech-2017 Conference, Stockholm)**. Prepared contents of pamphlets containing information about conference's sessions and timings, to be distributed in the conference. Assisted in setting up and wrapping up the infrastructure for conference in Stockholm University.
- **Cultural Stage Performances on India Day**, KTH-Stockholm, Nov 2017.
- **Participant (Ferienakademie 2016, Sarntal, Italy (18th to 30th September 2016))**. An entirely funded summer school in Sarntal, Italy. Sponsored by FAU Erlangen, TU Munich, and the University of Stuttgart for the participants. Participated in the course "Computer vision and human vision in daily life".
- **Active member (HAM Club at MIT, Manipal)**. Qualified the Amateur Station Operator's Certificate (Restricted Grade) Examination in 2013. Holds **Amateur Wireless Station License**, issued by The Ministry of Communications and IT, Government of India on 25/11/2014. The license grants the permission to establish and maintain a wireless telegraph transmitting and receiving station in India.
- **Active Member (National Cadet Corporation (NCC), Indian Military Cadet Corps (2010-2012))**. Along with undergraduate studies, I was an integral part of NCC for 2 years. **"B" and "C" Certificates** holder in NCC, under the Authority of Ministry of Defence, Government of India. Attended two **Combined Annual Training camps**, organized by NCC. Participation in events like tree plantation, clothes collection drive, organized by NCC. **Certificate of Appreciation** for Service in organizing a 1 day Event, including painting, lunch, inspirational movie display and park visits, for special care children, during the Golden Jubilee Celebration of 4 Kar Engr Coy, NCC, MIT, Manipal. **Represented MIT, Manipal** and won 1st Position - **The Best Contingent Award** in inter-college March-past competition held at Manipal University on **15th August (Independence Day)**, 2010 and also in 2011.
- **Management Committee Member (Robotics and Circuit Club at MIT, Manipal (2010-2011))**. Received **Certificate of Appreciation** - Recognized and recorded the outstanding service for being part of Robotics and Circuits Group and holding an exhibition, displaying different robots and circuits on 2/04/2011.